

IMPROVING SPEECH GENERATION WITH SHARED UNDERSTANDING AND LATENT RECONSTRUCTION

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ABSTRACT

Increasingly complex speech generation tasks motivate a shift from direct text-to-acoustic mapping toward understanding-informed synthesis. Speech understanding provides complementary knowledge through the reverse audio-to-text task, but parameter sharing alone does not directly supervise generated content. We propose a training framework that transfers this knowledge through shared learning and latent reconstruction feedback. Shared learning retains auxiliary text prediction on real audio while emphasizing generation. We further introduce Reconstruct CE, which reuses the native flow-training state and first velocity prediction to differentially reconstruct audio latents and applies transcript supervision through a frozen copy of the understanding branch. This connects local velocity regression with endpoint content supervision, without waveform decoding in the reconstruction branch or changes to generation-time decoding. Matched continuation experiments show that generation-emphasized shared learning reduces Chinese CER and English WER by 12.2% and 10.4% relative to generation-only continuation. In a separate matched-update ablation, Reconstruct CE further reduces CER/WER by 6.4%/8.0% relative to shared supervision, while improving predicted naturalness and preserving speaker similarity at the reported precision. The converged model achieves the highest speaker similarity among the evaluated systems in both languages under their respective generation settings.

Index Terms— speech generation, understanding supervision, flow matching, text reconstruction, multi-task learning

1. INTRODUCTION

As speech generation tasks become more complex, synthesis must jointly ensure faithful text realization, naturalness, and speaker preservation. This motivates moving beyond direct text-to-acoustic mapping toward understanding-informed generation. By learning the reverse audio-to-text relationship, speech understanding captures the correspondence between linguistic content and acoustic realization, providing complementary knowledge to guide synthesis.

Shared speech-text modeling offers one route for transferring this knowledge. SpeechT5 [1] learns shared representations through encoder-decoder pretraining, while UniVoice [2] combines autoregressive recognition with flow-based synthesis. Broader audio lan-

guage models extend joint learning across understanding and generation tasks [3, 4]. However, when text prediction is supervised on real audio, its influence on generation is mediated by shared parameters, without directly evaluating the content of reconstructed speech.

Feedback from synthesized audio provides a complementary route. The machine speech chain [5] reconstructs text from synthesized speech, StyleTTS 2 [6] uses speech-language-model representations for adversarial supervision, and Cont-SPT [7] incorporates ASR supervision into speech-tokenizer reconstruction. These studies demonstrate the value of speech-analysis feedback beyond auxiliary learning on real audio.

For flow-based generation, such feedback must connect local acoustic predictions with endpoint content. Flow matching [8], used in F5-TTS [9] and MMAudio [10], supervises local recovery velocities, whereas transcript supervision evaluates whether the reconstructed endpoint expresses the intended text. Differentiable reconstruction can bridge these objectives and propagate content gradients to the generator. This motivates our central question: **How can knowledge learned from speech understanding be effectively transferred to improve speech generation?**

We address this question through two complementary supervision paths. First, shared learning retains auxiliary text prediction on real audio during generation-focused training. Second, we introduce Reconstruct CE, which reuses the native flow-training state and first velocity prediction to differentially reconstruct speech latents, then applies transcript supervision through a frozen copy of the understanding branch. This connects local velocity regression with endpoint content supervision, without waveform decoding in the reconstruction branch or reader execution at inference. Matched bilingual experiments establish the benefits of shared supervision and frozen-reader feedback: the latter further improves content accuracy and predicted naturalness while preserving speaker similarity at the reported precision. Comparisons with six re-evaluated systems show that the converged model achieves the highest speaker similarity in both languages under the evaluated settings.

2. UNDERSTANDING SUPERVISION FOR GENERATION

Our framework uses understanding as shared supervision and reconstructed-content feedback (Fig. 1). A common latent interface lets the learned audio-to-text mapping read generated endpoints directly. Freezing this mapping supplies a fixed criterion; gradients through its input and the reconstruction trajectory update the generator.

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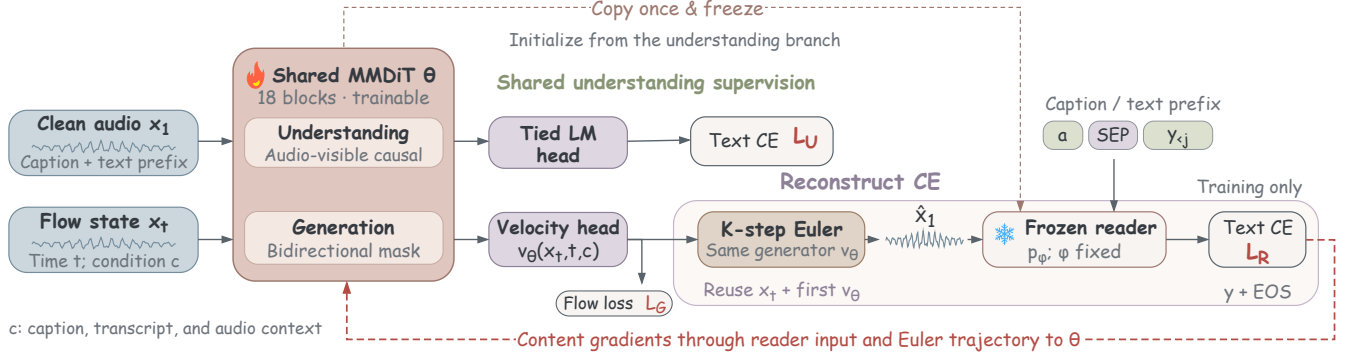


Fig. 1. Understanding supervision for speech generation. Shared learning combines flow matching with text prediction on real audio. Reconstruct CE reuses the flow-training state and first velocity for K differentiable Euler steps. Transcript gradients pass through a frozen reader’s input and the trajectory to the generator. Reconstruction is training-only. a/y : caption/transcript; waveforms represent audio latents.

2.1. Problem Formulation and Backbone

Let x_1 , y , and a denote clean audio latents, transcript, and caption. Generation conditions on $c = [a, \text{SEP}, y]$ and available audio context. Our 1.69B-parameter MMDiT has 18 joint blocks, width 1024, and 64-dimensional latents. The tasks share the backbone, embeddings, and projections, with a tied text output head. Following UniVoice [2], generation uses bidirectional audio–text attention; understanding restricts audio to audio and text to audio plus preceding text, preventing future-text leakage.

2.2. Shared Understanding Supervision

For Gaussian noise ϵ and $t \sim \mathcal{U}(0, 1)$, flow matching uses $x_t = tx_1 + (1 - t)\epsilon$. The generation loss $\mathcal{L}_G$ is the mean squared error between $v_\theta(x_t, t, c)$ and $x_1 - \epsilon$, averaged over valid predicted frames and latent dimensions. Known audio context is fixed to clean latents and excluded from this loss.

Shared text prediction trains the generation backbone to recover linguistic content, retaining audio-to-text supervision as generation receives greater weight. At $t = 1$, the model predicts text from clean audio. The understanding loss $\mathcal{L}_U$ is teacher-forced cross-entropy, with caption, separator, transcript, and EOS tokens as targets. We combine the objectives as $\mathcal{L}_{\text{joint}}^{(r)} = (r\mathcal{L}_G + \mathcal{L}_U)/(r + 1)$. We compare fixed and generation-emphasized ratios r ; reconstruction starts from a separate joint checkpoint (Sec. 3.1).

2.3. Differentiable Latent Reconstruction

Reconstruct CE extends the sampled flow-training state to a differentiable endpoint. With $z_0 = x_t$ and $h = (1 - t)/K$, we obtain $\hat{x}_1 = z_K$ through K differentiable Euler steps:

$$z_{k+1} = \Pi_c[z_k + h v_\theta(z_k, t + kh, c)], \quad (1)$$

for $k = 0, \dots, K - 1$. Here, Π_c fixes known audio context and clears padding at every step.

Reusing the exact first velocity ties endpoint transcript supervision to the same prediction trained for local recovery; subsequent Euler steps remain differentiable. Reconstruction reuses conditioning-dropout decisions without classifier-free guidance (CFG).

2.4. Frozen-Reader Content Supervision

We initialize reader p_ϕ once from the reconstruction run’s joint checkpoint and freeze it. Freezing preserves the learned audio-to-text mapping as a fixed content criterion. Differentiable reader inputs let transcript gradients update the generator through the Euler trajectory. At $t = 1$, it reads $\hat{x}_1$, including reconstructed and fixed-context regions:

$$\mathcal{L}_R = -\frac{1}{N_y} \sum_{j \in \mathcal{T}_y} \log p_\phi(y_j | \hat{x}_1, a, y_{<j}), \quad (2)$$

where $\mathcal{T}_y$ contains transcript and EOS positions and N_y counts valid target tokens. Caption and separator tokens provide context. Samples with dropped text conditioning contribute to flow matching but are excluded from $\mathcal{L}_R$.

We compute reader-input gradients at a detached endpoint leaf and transfer them to the original endpoint through a dot-product surrogate, preserving the chain-rule gradient.

Reconstruction training optimizes

$$\mathcal{L}_{\text{recon}} = \frac{10}{12} \mathcal{L}_G + \frac{1}{12} \mathcal{L}_U + \frac{1}{12} \mathcal{L}_R. \quad (3)$$

CE-off removes only $\mathcal{L}_R/12$, keeping the other coefficients fixed; the unfrozen-reader ablation also updates ϕ . The reconstruction branch needs no waveform decoding and runs only during training, leaving generation inference unchanged.

3. EXPERIMENTAL SETUP

3.1. Data and Controlled Training

The training mixture has 26.23M entries including replay, with Chinese/English proportions of 60/40. We use caption-plus-transcript conditioning, random-span inpainting, and text-conditioning dropout of 0.2. Seed-TTS-eval [11] contains 2020 Chinese and 1088 English cases. Our controlled configurations share reference audio, transcripts, and assignments of two fixed captions per language.

Two controlled continuations separate shared understanding from reconstruction feedback. Table 1B starts all branches from Joint 10:1 300k and adds 200k updates (500k total). Generation-only continuation removes the understanding objective, Joint 10:1 retains it, and Joint 20:1 increases the generation-to-understanding weight.

Table 1. Seed-TTS-eval. A: converged Reconstruct CE (800k+300k) and external systems; B: matched continuations (300k+200k). Ours: three inference seeds; baselines: one output/case. Ratios: generation/understanding. Bold: best/ties per block; -: unmeasured. MOS: 20 complete anonymous sessions (4/language), blinded randomized 1–5 quality ratings, 4 shared utterances from 20/language; 16 ratings/cell.

Model / setting	Params (B)	Chinese				English			
		CER (%) ↓	SIM ↑	UTMOS ↑	MOS ↑	WER (%) ↓	SIM ↑	UTMOS ↑	MOS ↑
<i>A. Comparison with re-evaluated speech models</i>									
Qwen2.5-Omni-7B	10.73	2.74	0.75	3.53	4.50	5.60	0.63	4.19	4.06
MiMo-Audio-7B-Instruct	8.02	3.10	0.58	2.66	3.06	4.03	0.44	3.54	3.13
Step-Audio 2-mini	8.32	1.41	0.64	3.51	3.88	1.89	0.47	4.31	3.75
UniAudio 2.0	4.86	5.63	0.48	3.09	3.00	4.76	0.34	3.84	3.31
Ming-Lite-Omni	0.69 (Talker only)	2.05	0.61	3.73	3.19	5.62	0.45	4.24	3.75
Ming-UniAudio-16B-A3B	18.27	1.06	0.70	2.77	3.56	1.82	0.58	3.67	3.44
Reconstruct CE (converged)	1.69	1.69	0.77	3.35	3.75	3.02	0.68	3.94	3.88
<i>B. Controlled continuations with shared supervision</i>									
Generation-only continuation	1.69	2.22	0.73	3.12	–	4.23	0.67	3.76	–
Joint 10:1	1.69	2.00	0.74	3.18	–	3.83	0.67	3.80	–
Joint 20:1	1.69	1.95	0.74	3.20	–	3.79	0.68	3.80	–

Table 2. Matched-update reconstruction: Joint 10:1 800k EMA +100k, $K = 10$, three inference seeds. CE-off removes only $\mathcal{L}_R/12$.

Training setting	Chinese			English		
	CER (%) ↓	SIM ↑	UTMOS ↑	WER (%) ↓	SIM ↑	UTMOS ↑
CE-off (shared supervision)	1.88	0.77	3.18	3.39	0.67	3.86
Reconstruct CE (unfrozen reader)	1.86	0.77	3.16	3.52	0.67	3.85
Reconstruct CE (frozen reader)	1.76	0.77	3.27	3.12	0.67	3.90

Table 2 starts all variants from Joint 10:1 800k EMA and adds 100k updates ($K = 10$; 900k total). CE-off retains the same flow and understanding loss coefficients; reconstruction variants add transcript supervision with a trainable or frozen reader. Table 1A reports the converged frozen-reader model: that 800k initialization plus 300k updates (1.1M total).

3.2. Baseline Evaluation

We evaluate released Qwen2.5-Omni-7B [12], MiMo-Audio-7B-Instruct [13], Step-Audio-2-mini [14], UniAudio 2.0 [3], Ming-Lite-Omni [15], and Ming-UniAudio-16B-A3B [16]. Qwen uses fixed Chelsie without reference conditioning, transcript-constrained greedy Thinker decoding, and a 4096-token Talker limit. MiMo uses reference-prompt TTS with a voice-matching instruction (global/local temperatures 0.6/0.9; top- p 1.0/0.95). Step uses temperature 0.7, a 4096-token limit, and reference-conditioned token-to-wave decoding. Ming-Lite uses its Talker/zero-shot frontend with reference speech, transcript, and speaker features. UniAudio uses `llm_ep2/ReasoningCodec` and `prompt_instruct_tts`, reference tokens, target text, and “Natural speech with a neutral speaking style.” (temperature 0.9, top- k 50, no CFG, 10 codec steps). Ming-UniAudio uses native text normalization, reference audio/transcript, 10 diffusion iterations, CFG=2, and 16-kHz output.

Baselines produce one output per case, with utterance-specific seeds derived from 42. Ming-Lite has 1086 English outputs after two decoder failures; all other counts are complete. UniAudio retries one invalid-codec case with another seed, without exclusions. Evaluators are shared, but generation interfaces differ; Qwen’s SIM is not a matched voice-cloning comparison.

Parameter counts exclude standalone codecs/vocoders and our training reader, but include Qwen’s integrated token-to-wave module. Ming-Lite counts only its Talker; MoE counts all experts; tied weights

count once.

3.3. Generation and Metrics

Our model uses EMA weights and per-utterance sampling with $v_{\text{cfg}} = (1 + g)v_{\text{cond}} - gv_{\text{uncond}}$, $g = 4$, and zeroed reference in the unconditional branch. Chinese/English use 48/64 Euler steps, time shift 1, and a common VAE posterior-sampling procedure for reference latents. Chinese CER uses Paraformer [17]; English WER uses Whisper large-v3 [18] through *faster-whisper*. SIM uses WavLM-Large features with ECAPA-TDNN [19] (`wavlm_large_finetune.pth`) and cosine similarity to paired reference audio; UTMOS [20] predicts naturalness. Automatic metrics are utterance means. Our Tables 1–2 average three inference seeds per trained checkpoint.

3.4. Auxiliary Protocols

Historical curves use learning rate 5×10^{-4} , seed 42, dynamic batches, and 2048 understanding tokens. Generation Only and Joint 10:1 load compatible weights from one source, but new text parameters may differ; Joint 20:1 restarts at Joint 10:1 300k. The historical runs describe learning trajectories; Tables 1B and 2 provide matched comparisons. The pilot uses CFG=4, batches of five, preserved punctuation, and historical reference conditioning. The pilot differs from unguided training reconstruction ($K = 10$) and main inference (48/64 steps).

4. RESULTS AND DISCUSSION

Shared understanding benefits generation. Table 1B isolates understanding retention from a common Joint 10:1 checkpoint. With 200k additional updates, Joint 10:1 reduces Chinese CER/English WER from 2.22%/4.23% to 2.00%/3.83% relative to generation-only continuation. Joint 20:1 reaches 1.95%/3.79%, corresponding to relative

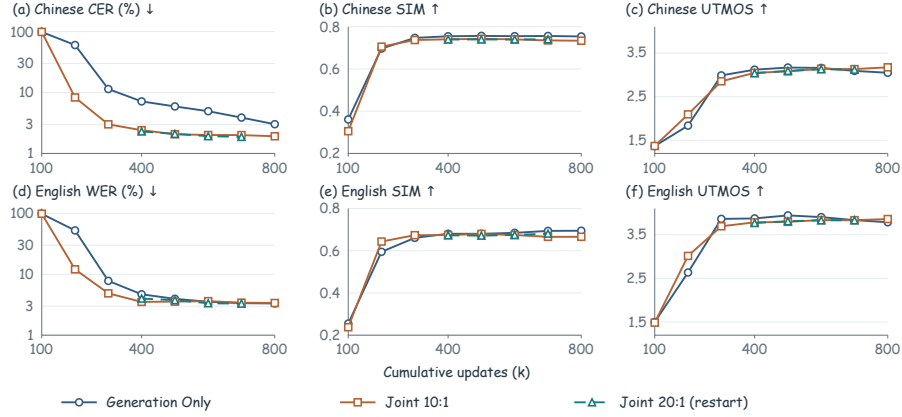


Fig. 2. Learning trajectories supplement Table 1B (seed 42; 100k intervals; logarithmic CER/WER axes). Joint 20:1 starts from Joint 10:1 300k and resets optimizer/data progress; the axis includes initialization updates (Sec. 3).

Table 3. Pilot: Joint 800k EMA, CFG=4 (two branches/step); *: selected depth. Errors (%). ZH/EN: Chinese/English.

Euler steps	4	5	8	10*	12	14	16
ZH CER ↓	3.57	2.65	1.98	1.89	1.89	1.85	1.88
ZH SIM ↑	0.67	0.69	0.73	0.73	0.73	0.73	0.73
ZH UTMOS ↑	2.47	2.95	3.20	3.23	3.24	3.24	3.24
EN WER ↓	3.92	3.58	3.47	3.41	3.29	3.35	3.31
EN SIM ↑	0.60	0.62	0.62	0.63	0.62	0.62	0.62
EN UTMOS ↑	3.49	3.73	3.91	3.92	3.94	3.95	3.94

reductions of 12.2%/10.4%, while increasing UTMOS from 3.12/3.76 to 3.20/3.80. With initialization and updates matched, these bilingual gains support maintaining understanding during generation-focused training. They are consistent with auxiliary text prediction helping shared parameters retain information useful for realizing linguistic content.

Reconstruction adds endpoint supervision. Table 2 evaluates Reconstruct CE from a separate shared-learning checkpoint under matched 100k-update continuations. Relative to CE-off, the frozen-reader variant reduces CER/WER from 1.88%/3.39% to 1.76%/3.12%, yielding 6.4%/8.0% relative reductions. CE-off already retains ordinary understanding supervision, and the coefficients of the flow and understanding losses are unchanged. The comparison therefore supports the value of supervising reconstructed content beyond auxiliary prediction on real audio. Whereas flow matching constrains a local velocity prediction, Reconstruct CE evaluates transcript consistency at the integrated endpoint and propagates that signal through the reconstruction trajectory. The observed gains are consistent with these two objectives supplying complementary supervision.

Freezing strengthens content feedback while preserving speech fidelity. Freezing reduces CER/WER from 1.86%/3.52% to 1.76%/3.12% compared with updating the reader jointly. Mechanistically, freezing holds the audio-to-text mapping fixed, so the reconstruction objective acts on the generator through the reader input rather than also changing the reader’s parameters. External waveform recognizers measure the accuracy gains, demonstrating benefits beyond the training reader itself. Relative to CE-off, frozen reconstruction also raises UTMOS from 3.18/3.86 to 3.27/3.90, with SIM unchanged at 0.77/0.67 at the reported precision. Thus, the

added content objective improves predicted naturalness alongside transcription accuracy while retaining speaker similarity. These benefits require the reader only during training and leave generation-time decoding unchanged.

Converged content accuracy and speaker preservation. After 300k reconstruction updates from the 800k joint checkpoint, the model achieves 1.69% CER and 3.02% WER (Table 1A). Its SIM (0.77/0.68) is highest among the evaluated systems, and all three automatic measures improve over MiMo-Audio and UniAudio 2.0 in both languages. Compared with reference-conditioned Ming-UniAudio, it achieves higher SIM (0.77/0.68 versus 0.70/0.58) and UTMOS (3.35/3.94 versus 2.77/3.67). Under the interfaces in Sec. 3.2, these results characterize the final system’s content accuracy and speech fidelity, complementing the controlled evidence for its supervision design.

Learning trajectories and sampling depth. Historical trajectories (Fig. 2) show earlier error reductions with joint learning, consistent with Table 1B. In the separate pilot (Table 3), 4 to 10 Euler steps reduce CER/WER from 3.57%/3.92% to 1.89%/3.41%, while SIM and UTMOS approach relatively stable ranges around 8–10 steps. These trends complement the matched evidence with learning and sampling behavior.

5. CONCLUSION

We introduced shared learning and latent reconstruction feedback to improve generation through speech understanding. Shared text prediction remains beneficial when generation receives greater training emphasis, while Reconstruct CE adds direct transcript supervision to the generator’s reconstructed endpoint. By freezing the latent reader and preserving gradients through its input and the reconstruction trajectory, the method supplies a fixed content criterion alongside local flow matching. Controlled bilingual experiments support the complementary roles of these supervision paths, with improvements in transcription accuracy and predicted naturalness while retaining speaker similarity at the reported precision. The converged model demonstrates their practical value. These findings establish understanding as both an auxiliary learning task and a source of content supervision that leaves generation-time decoding unchanged. Future work will extend feedback to prosody and emotion.

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